Week 3

Case Study 1

Data for Questions 1 to 5

Tutoring Session Readiness Score

An online tutoring platform uses a small feedforward network to estimate a session-readiness score before assigning a tutor. The model uses two input features:

$$x_1 : \text{student preparation score}, \quad x_2 : \text{topic difficulty score}.$$

For one session, $x_1 = 2$ and $x_2 = 1$. The network computes hidden quantities first and then combines them:

$$z_1 = 2x_1 + x_2, \qquad z_2 = x_1 - x_2 + 1,$$

$$\hat{y} = z_1 + 2z_2.$$

The computation is feedforward: inputs are used to compute z_1, z_2, and these hidden quantities are then used to compute $\hat{y}$.

Question 1: What is the value of z_1 for this session?

A) 6

B) 5

C) 4

D) 3

Correct Answer: B

Explanation: $z_1 = 2x_1 + x_2 = 2(2) + 1 = 5$.

Question 2: What is the value of z_2?

A) 0

B) 1

C) 3

D) 2

Correct Answer: D

Explanation: $z_2 = x_1 - x_2 + 1 = 2 - 1 + 1 = 2$.

Question 3: What is the final value of $\hat{y}$?

A) 9

B) 7

C) 8

D) 6

Correct Answer: A

Explanation: $\hat{y} = z_1 + 2z_2 = 5 + 2(2) = 9$.

Question 4: Which computation must be completed before $\hat{y}$ is evaluated?

A) The loss must be minimized over all stored tutoring sessions.

B) The target value must be changed by the model before scoring.

C) The hidden quantities z_1 and z_2 must be computed.

D) The learning rate must be set to zero.

Correct Answer: C

Explanation: The output depends directly on z_1 and z_2, so they must be computed first.

Question 5: What makes this a feedforward computation?

A) The output is repeatedly fed back as a new input.

B) Information flows from inputs to hidden quantities to output.

C) The target label is used before the input features.

D) The network stores a separate state for each previous session.

Correct Answer: B

Explanation: In a feedforward network, values are computed in one direction from inputs through intermediate units to the output.

Case Study 2

Data for Questions 6 to 10

Inventory Forecast Calculation Review

A store wants to understand how a change in one product-level input can affect a final forecasting loss. Before training the full inventory model, the analytics team writes a small computational graph on a whiteboard and follows both the forward values and the local derivatives. For one product, the graph is

$$u = 2x, \qquad v = u + 3, \qquad L = v^2.$$

For the current product, $x = 1$. The team computes local derivatives using the chain rule. Here, x is a simplified demand signal, u and v are intermediate quantities, and L is the scalar loss that would be reduced during training.

Question 6: What is the value of u?

A) 2

B) 1

C) 3

D) 4

Correct Answer: A

Explanation: Since $u = 2x$ and $x = 1$, $u = 2$.

Question 7: What is the value of v?

A) 3

B) 4

C) 6

D) 5

Correct Answer: D

Explanation: $v = u + 3 = 2 + 3 = 5$.

Question 8: What is the loss L?

A) 20

B) 15

C) 25

D) 10

Correct Answer: C

Explanation: $L = v^2 = 5^2 = 25$.

Question 9: What is $\frac{\partial L}{\partial v}$ at $v = 5$?

A) 5

B) 10

C) 25

D) 2

Correct Answer: B

Explanation: Since $L = v^2$, $\frac{\partial L}{\partial v} = 2v = 10$.

Question 10: What is $\frac{\partial L}{\partial x}$?

A) 10

B) 5

C) 40

D) 20

Correct Answer: D

Explanation: By the chain rule, $\frac{\partial L}{\partial x} = \frac{\partial L}{\partial v}\frac{\partial v}{\partial u}\frac{\partial u}{\partial x} = 10(1)(2) = 20$.

Case Study 3

Data for Questions 11 to 15

Sensor Calibration Error Analysis

A lab calibrates a sensor whose raw reading must be converted into a corrected measurement. To check the backpropagation calculations before using many sensor logs, the engineers use a one-hidden-unit model and one training example. For this example,

$$x = 2, \qquad y = 5.$$

The model is

$$h = w_1 x + b, \qquad \hat{y} = w_2 h,$$

with $w_1 = 1,\ b = 0,\ w_2 = 2$. The loss is

$$L = \frac{1}{2}(\hat{y} - y)^2.$$

Backpropagation computes gradients from the loss back through the output and hidden computations. The goal is not to build a production sensor model from one example, but to verify how the prediction error is assigned to w_2, then passed backward through h to w_1.

Question 11: What is the hidden value h?

A) 1

B) 4

C) 2

D) 5

Correct Answer: C

Explanation: $h = w_1 x + b = 1(2) + 0 = 2$.

Question 12: What is the prediction $\hat{y}$?

A) 3

B) 5

C) 2

D) 4

Correct Answer: D

Explanation: $\hat{y} = w_2 h = 2(2) = 4$.

Question 13: What is $\frac{\partial L}{\partial \hat{y}}$?

A) -1

B) 1

C) -2

D) 2

Correct Answer: A

Explanation: $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y = 4 - 5 = -1$.

Question 14: What is $\frac{\partial L}{\partial w_2}$?

A) -1

B) -2

C) 2

D) 1

Correct Answer: B

Explanation: Since $\hat{y} = w_2 h$, $\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} h = (-1)(2) = -2$.

Question 15: What is $\frac{\partial L}{\partial w_1}$?

A) -1

B) 1

C) -4

D) 4

Correct Answer: C

Explanation: $\frac{\partial L}{\partial h} = (-1)w_2 = -2$, and $\frac{\partial h}{\partial w_1} = x = 2$, so $\frac{\partial L}{\partial w_1} = -4$.

Case Study 4

Data for Questions 16 to 20

Course Recommendation Score Update

A course platform tracks whether a learner is likely to engage with a recommended module. For an internal training exercise, the team uses a simple linear prediction model so that each gradient and parameter update can be inspected. The model predicts a learner's interest score as

$$\hat{y} = w_1 x_1 + w_2 x_2 + b.$$

For one learner,

$$x_1 = 1, \quad x_2 = 2, \quad y = 6.$$

Initially, $w_1 = 1,\ w_2 = 1,\ b = 0$. The loss is

$$L = \frac{1}{2}(\hat{y} - y)^2.$$

The learning rate is $\eta = 0.1$, and parameters are updated using

$$w \leftarrow w - \eta \frac{\partial L}{\partial w}.$$

The two inputs summarize recent activity and topic match, while the bias accounts for a baseline interest level. The questions focus on how one observed error changes the weights and bias after a gradient descent update.

Question 16: What is the initial prediction $\hat{y}$?

A) 4

B) 3

C) 5

D) 2

Correct Answer: B

Explanation: $\hat{y} = 1(1) + 1(2) + 0 = 3$.

Question 17: What is $\frac{\partial L}{\partial \hat{y}}$?

A) -3

B) 3

C) -6

D) 6

Correct Answer: A

Explanation: $\frac{\partial L}{\partial \hat{y}} = \hat{y} - y = 3 - 6 = -3$.

Question 18: What is $\frac{\partial L}{\partial w_2}$?

A) 3

B) -3

C) 6

D) -6

Correct Answer: D

Explanation: $\frac{\partial L}{\partial w_2} = (\hat{y} - y)x_2 = (-3)(2) = -6$.

Question 19: What is the updated value of w_1?

A) 0.7

B) 1.6

C) 1.3

D) 0.4

Correct Answer: C

Explanation: $\frac{\partial L}{\partial w_1} = (-3)(1) = -3$, so $w_1 = 1 - 0.1(-3) = 1.3$.

Question 20: What is the updated value of b?

A) 0.3

B) -0.3

C) 0.6

D) -0.6

Correct Answer: A

Explanation: $\frac{\partial L}{\partial b} = -3$, so $b = 0 - 0.1(-3) = 0.3$.